

Black-Litterman Model: Definition, Basics, and Example

 [investopedia.com/terms/b/black-litterman_model.asp](https://www.investopedia.com/terms/b/black-litterman_model.asp)

What Is the Black-Litterman Model?

The Black-Litterman model is an analytical tool used by portfolio managers to optimize asset allocation within an investor's risk tolerance and market views. The Black-Litterman model starts from a neutral position using modern portfolio theory (MPT), and takes additional input from investors' views to determine how the ultimate asset allocation should deviate from the initial portfolio weights. It then undergoes the process of mean-variance optimization (MVO) to maximize the expected return given one's objective risk tolerance.

Key Takeaways

- The Black-Litterman model is a portfolio allocation model that begins with modern portfolio theory and adds investor views of expected returns.
- The MPT model is considered limited as it only incorporates historical market data and then assumes those same returns going forward.
- The BL model lets the investor apply their own views to it, and then optimizes the recommended asset allocation.

Understanding the Black-Litterman Model

The Black-Litterman model for portfolio construction is based on MPT. Modern portfolio theory posits that an investment's risk and return characteristics should not be viewed alone, but should be evaluated by how the investment affects the overall portfolio's risk and return. MPT shows that an investor can construct a portfolio of multiple assets that will maximize returns for a given level of risk.

Given a desired level of expected return, an investor can similarly construct a portfolio with the lowest possible risk. Based on statistical measures such as variance and correlation, an individual investment's performance is less important than how it impacts the entire portfolio. The BL model was designed to improve on this model since one of the limitations of MPT is that it assumes that past expected returns will continue in the future.

Other pricing models, such as the capital asset pricing model (CAPM), may produce different expectations from past performance. The BL model incorporates observed market data along with an investor's projections of future expected returns based on models like CAPM or others. The model essentially modifies the default MPT allocation by taking into account expectations of future performance.

The BL approach allows any model estimation error to become apparent as allocation choices may magnify poor assumptions.

Special Considerations

The BL model was created in 1990 by Goldman Sachs economists Fischer Black (of Black-Scholes model fame) and Robert Litterman. The aim was to develop "a quantitative and disciplined approach to structuring international bond portfolios in a manner consistent with the portfolio manager's unique view of markets."¹ It continues to be used and is recognized as a key mathematical tool in the investment world.

The goal of the BL model is to improve MPT-generated asset allocations by incorporating opinions on future outlook. While these projections are merely opinions or the result of pricing models that rely on subjective inputs, the BL model may result in bias or incorrect assumptions.

For instance, an overly optimistic view of one asset class will result in having a greater portfolio weight than MPT would recommend. If that asset class falters, it can result in magnified losses. Investors who use the Black-Litterman model should be aware of this and regularly update their expectations, rebalancing their portfolio weights accordingly.

The Black-Litterman model can help global investors, such as pension funds and insurance companies, decide how to allocate their investments across different asset classes and countries.

Advantages and Disadvantages of the Black-Litterman Model

As with any other investment strategy, there are benefits and drawbacks that investors must weigh out before using the Black-Litterman model. We've highlighted some of the most common pros and cons below.

Advantages

One advantage of the model is that it uses the mean-variation approach. This method allows investors to take their risk tolerance into account so they can maximize their expected returns.

Investors can also use their own views when using this tool, which gives them more control and confidence over how their portfolio is structured in terms of asset allocation and how it's rebalanced when it comes time to make any changes.

Furthermore, it is more instinctive, which helps prevent heavy changes in portfolio weight over time.

Disadvantages

One of the main drawbacks of using the Black-Litterman model is that it doesn't promise or guarantee the best portfolio. Rather, it devises a portfolio (and rebalances it if/when needed) based on the investor's or portfolio manager's views about the market.

Another key disadvantage is that since it works on assumptions, it is sensitive to the investor's views. This means that the model is based on the fact that these views act independently of one another and, therefore, don't act together.

Pros

- Mean-variation approach allows investors to factor in risk tolerance to maximize returns
- Investors can input their own views when using the tool
- More instinctive and intuitive, preventing heavy changes in portfolio weight

Cons

- Doesn't guarantee the best portfolio
- Works on assumptions

Example of the Black-Litterman Model

Let's assume that a portfolio management team at an insurance company is extremely bullish on developing country markets in the year ahead.

The initial asset allocation to emerging markets from modern portfolio theory is 10%. After confirming their opinions with various pricing models and economic outlooks for the region, they are inclined to be overweight emerging markets stocks.

After putting this bullish view into the BL model, they perform MVO and allow their portfolio to contain emerging markets securities of up to 15%.

Who Created the Black-Litterman Model?

The Black-Litterman model was created by Fischer Black and Robert Litterman in 1990. Both men were economists at Goldman Sachs. Their goal was to develop a disciplined method that would allow portfolio managers to structure investment portfolios that aligned with their market views.¹ The model is still used today to create and rebalance portfolios.

What Is Portfolio Optimization?

Portfolio optimization is the process or strategy of managing assets in an investment portfolio. It is also commonly known as modern portfolio theory. It involves choosing the investments you think would be suitable to maximize returns by assuming a certain level of risk. Portfolio optimization or MPT uses diversification as a way for investors to reach their goals. This is done by investing in different asset classes and securities, such as bonds, stocks, and mutual funds.

What Are the Main Pros and Cons of Using the Black-Litterman Model.

The Black-Litterman model is a tool used by investors and portfolio managers to create and review financial portfolios.

One of the main advantages is that it allows market views to be taken into account when crafting or changing portfolios. This allows the investor's risk tolerance to be factored in so returns can be maximized.

But, there are drawbacks. Because it relies on an investor's views and assumptions, there is no guarantee or promise that using the Black-Litterman model will guarantee returns.

The Bottom Line

Investors and portfolio managers use various tools to help them devise and make changes to investment portfolios. The Black-Litterman model is one. Created by two Goldman Sachs economists in 1990, the model is still used by investors and professionals. Although it allows you to input your personal views about the market, it isn't necessarily a foolproof tool. You should be aware that it doesn't guarantee returns.